World-Model Value Depends on the Policy That Acts

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OUR POSITION

Evaluate world models together with the policies that act on their predictions.

A policy with better baseline performance can gain more from the same advice.

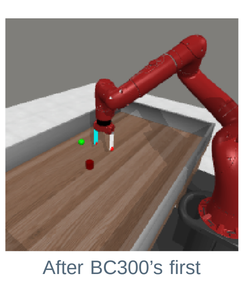
+11.50 mm

Change in advice gain
BC100 → BC300

01 Execute the same five actions; vary the policy used next.

Task: push the red object to the green goal (Meta-World [1]). Frozen predictor: JEPA-WM [2]. BC = behavior cloning.

Action advice = five robot commands selected using predicted outcomes. Each command controls hand motion and the gripper.



After BC300's first 10 actions

Same saved state → Steps 10–15

- Fixed BC300 actions → Test policy: BC100, BC300, ... → Baseline distance
- Fixed WM-selected actions → Test policy: BC100, BC300, ... → Distance with advice

Execute five stored actions → **Same test policy in both runs** → **Measure final distance**

Replan every five actions; steps 15–100

Average over steps 91–100

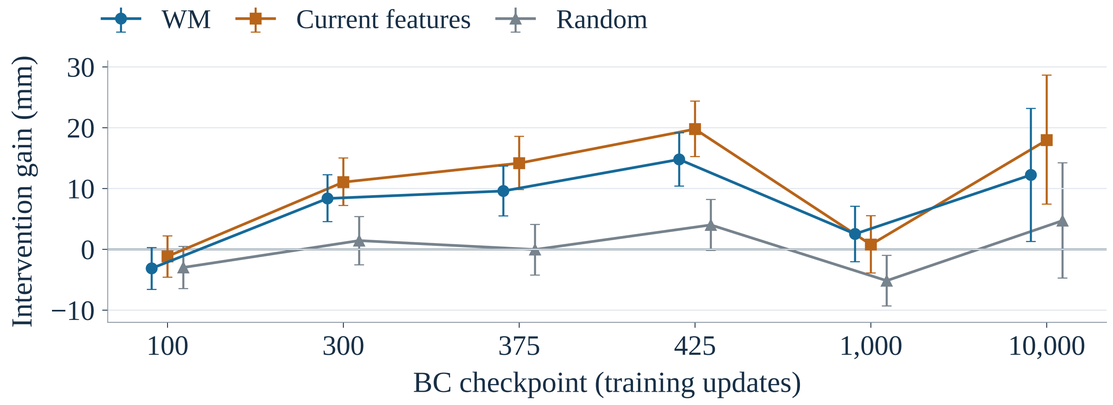
Gain = baseline − advice

Two roles: BC300 fixes the baseline plan. The test policy (possibly BC300) proposes 20 actions, executes five, and replans from its new state. The simulator state carries through the switch at step 15.

WM selection: a learned scorer evaluates nine 20-action forecasts; execute five actions. Across test policies, fix the step-10 state, candidates, forecasts, and both stored action sequences.

02 BC300 gains more from the same advice

512 shared states; one five-command intervention.
BC100/300 denote 100/300 training updates.



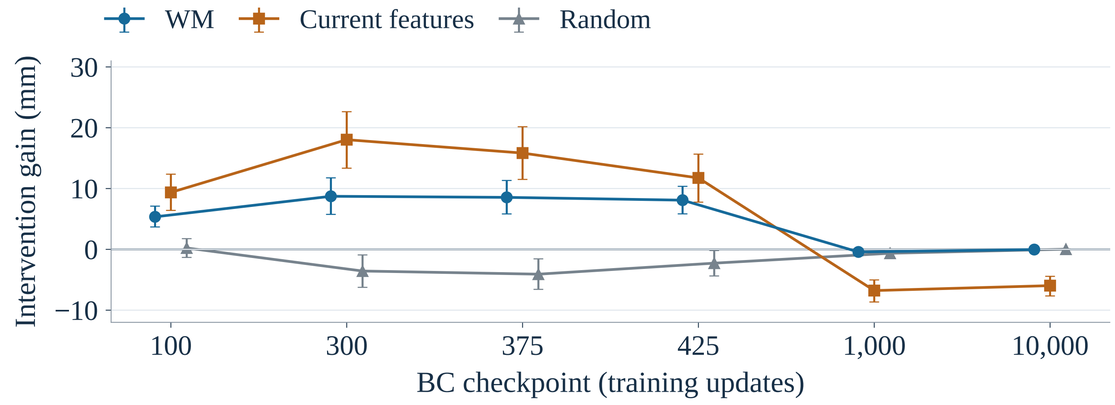
Final distance and gain (mm)

Policy	Baseline ↓	With advice ↓	Gain ↑
BC100	143.51	146.64	−3.13
BC300	121.08	112.71	**+8.37**

Baseline improves **22.43 mm**; advice gain rises **11.50 mm**.
Gain increase: paired 95% CI [6.80, 16.25] mm.

03 Current observations guide useful actions

1,024 cases; choose actions repeatedly during execution.
Baseline: the tested policy acts alone throughout.



+8.73 vs. +18.04 mm

WM vs. learned selection from current observations (BC300).

Current-observation selection delivers the larger gain.

Current-observation scorer inputs: current/goal features + candidate actions. Both scorers share the encoder, candidate rule, capacity and supervision; thresholds are validated separately.

Metric: gain = baseline − intervention distance, averaged over steps 91–100; positive is better. Bars: paired 95% intervals. Protocols differ in targets, thresholds, schedules, and baselines.
Fixed-advice evidence: BC100 → BC300 was fixed before fresh evaluation. Gain at BC300, gain increase, and baseline improvement have positive family-adjusted lower bounds (15 tests; α = .010).

04 Later policy control changes the outcome

From the same step-30 state, simulate nine 20-action plans.
Execute all 20, then let BC1000 finish each run. Choose by actual distance at step 50; compare with the original plan.

Step 50: closer to goal
+7.79 mm
95% CI [7.15, 8.48]

→

Task end: farther from goal
−3.68 mm
95% CI [−6.15, −1.74]

Evaluate selected plans through task completion.

Compared with the original 20-action plan + BC1000.
Post hoc: 256 separate cases; original BC inputs differ from the primary study.

05 Correct forecast pairing improves selection

Exploratory; 512 reused states; five-step forecasts; BC300.
Scorers also receive BC300's five actions at each of eight probe states.

Correct pairing	Shuffled pairing	Current features
17.08 mm	**5.48 mm**	**15.13 mm**

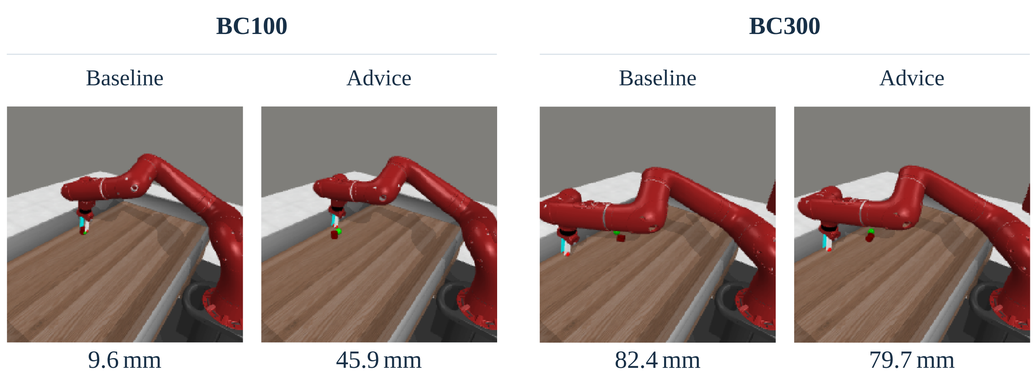
Matched − shuffled: **+11.60 mm** [7.37, 15.90].

Shuffled: reassign forecasts across plans in training and evaluation.
Matched − current: +1.95 mm [−0.96, 4.88]; the interval spans zero.

06 Robot examples: the same actions have different effects

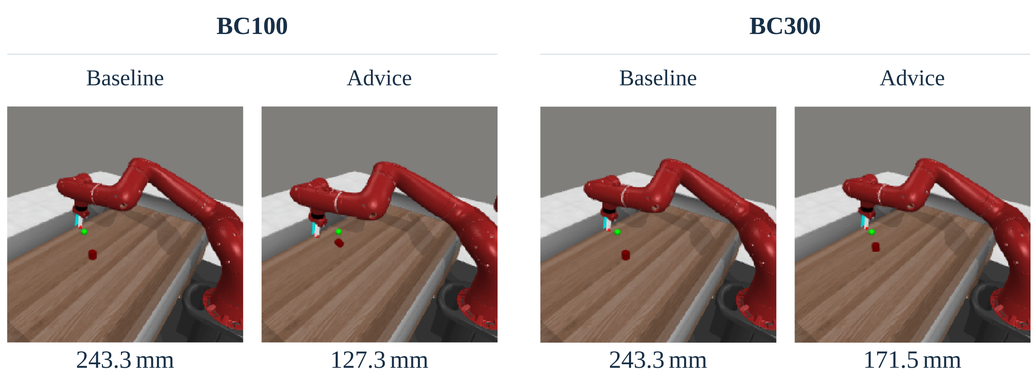
Separate exploratory examples; the scorer receives the same BC300 action description for both continuation policies.

(a) Advice changes from harmful to helpful



Gain: −36.3 → +2.6 mm (baseline worsens)

(b) Both policies benefit, by different amounts



Gain: +116.0 → +71.8 mm

Actual simulated states at step 100; labels average steps 91–100. Cases chosen after evaluation near each sign group's median gain change.
Across 512 cases: 252 increased / 177 decreased / 83 unchanged gains. Report the baseline alongside the advice gain for each policy.

Evaluate predictive benefit for each policy through task completion.

Report baseline and advice gain; compare current-observation selection; include inference cost.

One task/BC run; privileged observations; physics paused. BC300 candidate evaluation: WM 1.001 s vs. direct .057 s/episode.

[1] Yu et al., *Meta-World*, CoRL 2020. [2] Terver et al., *What Drives Success in Physical Planning with Joint-Embedding Predictive World Models?*, 2025.

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